

## **LAB NO: 09**

**LAB TITLE :** Introduction to Functions in C++

### **OBJECTIVE :**

The objective of this lab is to understand the basic concept of functions in C++. This lab helps students learn how to declare functions, create function prototypes, call functions from the main program, pass parameters to functions, and return values from functions. Functions make programs more organized, reusable, and easy to understand.

### **THEORY :**

A function is a block of code that performs a specific task. Functions are used to divide a large program into smaller parts. A function can be called many times in a program which reduces code repetition.

Every function has:

1. Function name
2. Return type
3. Parameters
4. Function body

Functions improve:

1. Code readability
2. Program organization
3. Reusability of code

### **Example**

```
int add(int a, int b);
```

### **Prototype of Function :**

The prototype of a function is a way of describing the parameters (also called as arguments of function) and parameter types with which a legal call to the function can be made. It contains the name of the function, its parameters and their type, and the return value.

### **Syntax :**

```
returnType functionName(parameters);
```

### **Function Definition :**

Function definition is the part of the program where the actual work of the function is written. It contains the statements and instructions that the function performs.

### **Syntax :**

```
returnType functionName(parameters)
{
    // body of function
}
```

### **Function Call :**

Function call is used to execute or run a function in the program. When a function is called, the control goes to the function and performs the required task.

### **Syntax :**

```
functionName();
```

### **SOFTWARE :**

1. DEV-C++ IDE.
2. C++ Compiler.

### **LAB TASKS :**

#### **Function Introduction and Declaration.**

1. **Declare a function called "greet" that takes no parameters and returns nothing.**
2. **Inside the function, display a greeting message of your choice.**
3. **Call the function from the main program to execute the greeting.**

```
#include <iostream>
using namespace std;
```

```

void displayGreeting(); // Function prototype

int main()
{
    displayGreeting(); // Function call karna

    return 0;
}

void displayGreeting() // Function definition
{
    cout << "Welcome to C++ Programming"; // Greeting message display karna
}

```

### Basic Function Declaration and Call

1. Declare a function called "sayHello" that takes no parameters and returns nothing.
2. Inside the function, display the message "Hello, World!".
3. Call the function from the main program to execute the greeting.

```

#include <iostream>
using namespace std;

void sayHello(); // function prototype

int main()
{
    sayHello(); // function ko call karna

    return 0;
}

void sayHello() // function definition
{
    cout << "Hello, World!"; // message print karna
}

```

**Write a C++ program that: Takes 10 integer values as input from the user and stores them in a 1D**

array for. Creates a function named calculateSum that: Accepts the array and its size as parameters. Returns the sum of all elements In the main() function: Call the function. Calculate and display the average of the array elements Constraints: Use only one-dimensional array Use function for sum calculation.

```
#include <iostream>
using namespace std;

int calculateSum(int arr[], int size); // function prototype

int main()
{
    int numbers[10];
    int sum;
    float average;

    for (int i = 0; i < 10; i++)
    {
        cin >> numbers[i]; // user se values lena
    }

    sum = calculateSum(numbers, 10); // function call karna
    average = sum / 10.0; // average calculate karna

    cout << "Sum = " << sum << endl;
    cout << "Average = " << average;

    return 0;
}

int calculateSum(int arr[], int size) // function definition
{
    int total = 0;

    for (int i = 0; i < size; i++)
    {
```

```
        total = total + arr[i]; // sum calculate karna
    }

    return total; // total return karna
}
```

## CONCLUSION

In this lab, we learned the concept of functions in C++. We studied function prototypes, function definitions, function calls, parameters, and return values. We also learned how functions make programs organized and reusable. This lab improved our understanding of modular programming in C++.